# 1 Chebyshev Binomial Integral

**Theorem.** Given an integral of the form

$$\int x^m (a + bx^n)^{p/q} dx$$

where $m, n, p/q \in \mathbb{Q}$, $a, b \in \mathbb{R}$, and $p/q$ is a reduced fraction.

1. **Case 1:** $p/q$ **is an integer.**
$$\frac{p}{q} \in \mathbb{Z}$$
Let $x = t^s$, where $s$ is the lcm (least common multiple) of the denominators of $m$ and $n$.

2. **Case 2:** $\frac{m+1}{n}$ **is an integer.**
$$\frac{m+1}{n} \in \mathbb{Z}$$
Let $a + bx^n = t^q$.

3. **Case 3:** $\frac{m+1}{n} + \frac{p}{q}$ **is an integer.**
$$\frac{m+1}{n} + \frac{p}{q} \in \mathbb{Z}$$
Let $ax^{-n} + b = t^q$ (or equivalently, $\frac{a+bx^n}{x^n} = t^q$).

If $a \neq 0$ and $b \neq 0$, then these three cases are the only cases where the result can be represented in elementary functions.

## Exercise 1

The integral to be evaluated is:

$$\int \frac{dx}{\sqrt[3]{x^2}(1+\sqrt[3]{x})}$$

Note that the integral can be rewritten in the form of a Chebyshev Binomial Integral:

$$\int \frac{dx}{\sqrt[3]{x^2}(1+\sqrt[3]{x})} = \int x^{-2/3}(1+x^{1/3})^{-1}dx$$

Comparing this with the general form $\int x^m(a+bx^n)^{p/q}dx$:

- $m = -2/3$
- $n = 1/3$
- $p/q = -1$ (which is an integer)
- $a = 1, b = 1$

## Exercise 2: Surface Area of Revolution

**Calculate the surface area of the solid of revolution formed by rotating the curve**

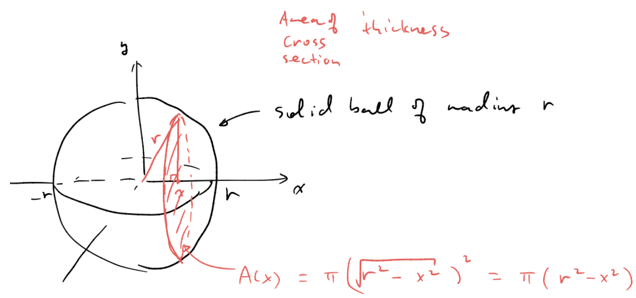
$$y = x^{3/4}$$

**on the interval $[0, 1]$ about the $x$-axis.**

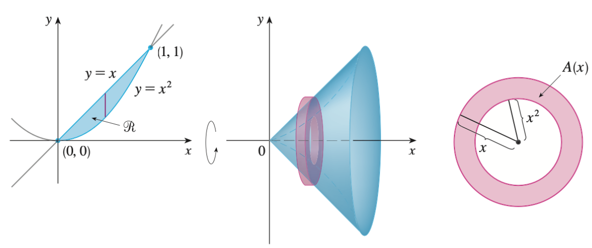
# 2 Volume Calculation

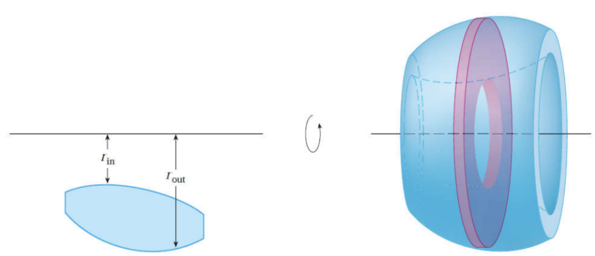
**Definition of Volume** Let $S$ be a solid that lies between $x = a$ and $x = b$. If the cross-sectional area of $S$ in the plane $P_x$, through $x$ and perpendicular to the $x$-axis, is $A(x)$, where $A$ is a continuous function, then the **volume** of $S$ is

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} A(x_i^*)\,\Delta x = \int_a^b A(x)\,dx$$




## 2.1 Washer Method



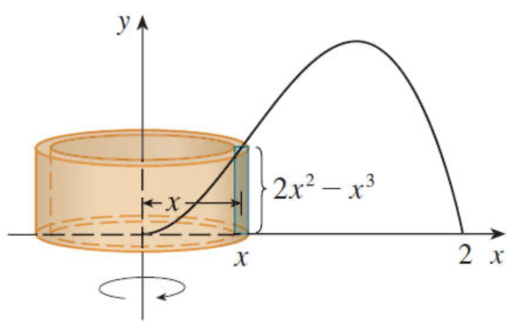

$$V = \pi \int_a^b \left(R^2 - r^2\right) dx \tag{1}$$

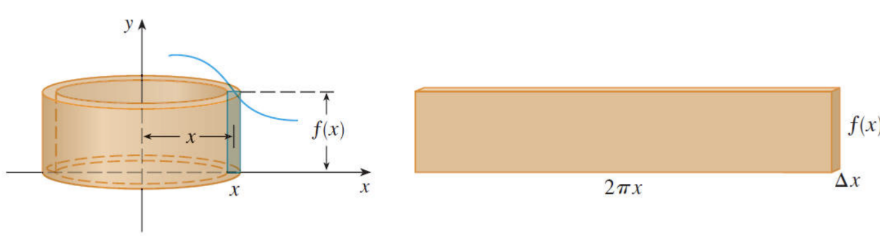
The volume can be regarded as a pile of slices, like building up a house.

**Scope of Application (Typical Scenarios):**

- Rotation about the $x$-axis, slicing parallel to the $y$-axis (i.e., integration with respect to $dx$).
- Rotation about the $y$-axis, slicing parallel to the $x$-axis (i.e., integration with respect to $dy$).

## 2.2 Shell/Cylindrical Method




$$V = 2\pi \int_a^b x \cdot f(x) \cdot dx \tag{2}$$

**Scope of Application (Typical Scenarios):**

1. Rotation about the x-axis, slicing parallel to the x-axis (i.e., integration with respect to $dy$)
2. Rotation about the y-axis, slicing parallel to the y-axis (i.e., integration with respect to $dx$)

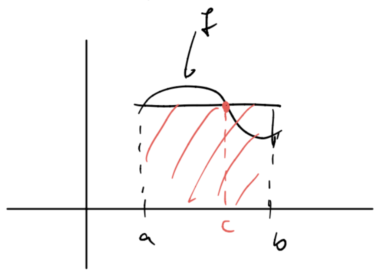
# 3 MVT for Integrals

Mean Valued Theorem for Integrals

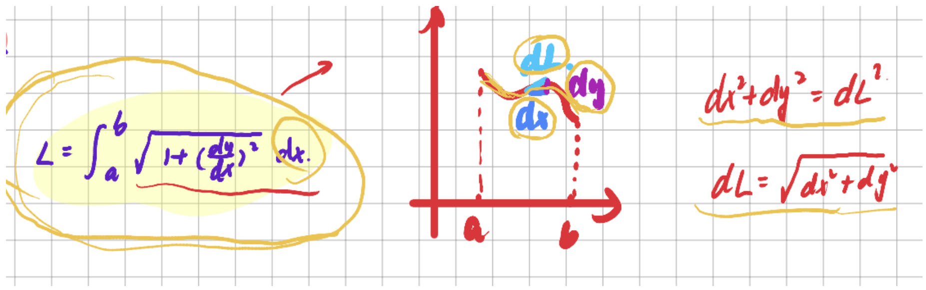
If $f$ is continuous on $[a,b]$, then $\exists\, c \in [a,b]$ s.t.

$$\int_a^b f(x)\,dx = f(c)\,(b-a)$$

$$\Leftrightarrow \quad \frac{1}{b-a}\int_a^b f(t)\,dt = f(c)$$



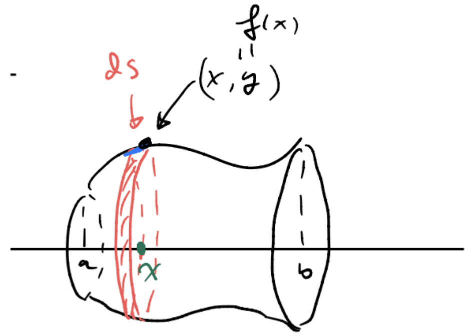
# 4 Arclength

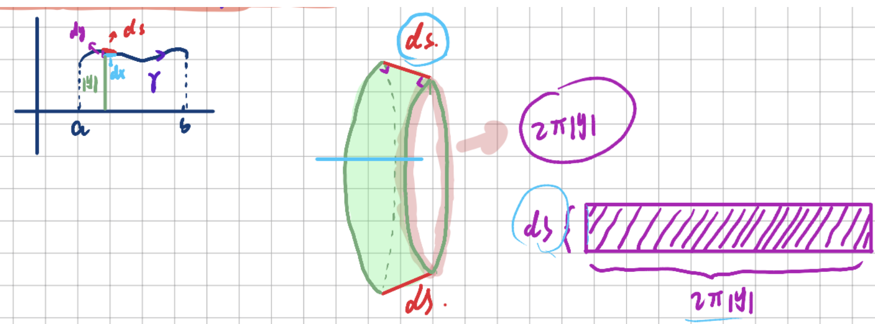



## 4.1 Arclength Function

$$S(x) = \int_a^x \sqrt{1 + \left(\frac{dy}{dt}\right)^2}\, dt \tag{3}$$

## 5 Area of Surface of Revolution




$$A = \int_{\gamma} 2\pi|y|\,ds$$

$$= \int_a^b 2\pi|y|\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$

$$= 2\pi\int_a^b |f(x)|\sqrt{1+f'(x)}\,dx$$

# 6 Parametric Curves

## 6.1 Slope

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\dot{y}(t)}{\dot{x}(t)} \quad \text{where } \dot{x}(t) = dx/dt \neq 0 \tag{4}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{\dot{y}(t)}{\dot{x}(t)}\right) = \frac{d}{dt}\left(\frac{\dot{y}(t)}{\dot{x}(t)}\right)\frac{dt}{dx} = \frac{d}{dt}\left(\frac{\dot{y}(t)}{\dot{x}(t)}\right)\frac{1}{\dot{x}(t)} = \frac{\ddot{y}\dot{x} - \ddot{x}\dot{y}}{(\dot{x})^3} \tag{5}$$

## 6.2 Tangent Line

1. If $y'(t) = 0, x'(t) \neq 0$, we have a horizontal tangent line.
2. If $y'(t) \neq 0, x'(t) = 0$, we have a vertical tangent line.
3. If $x'(t) = y'(t) = 0$, we need to use the definition of slope of tangents.

Example $r = 1 + sin\theta$ find the slope of tangent at $\theta = \frac{3\pi}{2}$

### 6.3 Arclength/Surface Area in Parametric Form

**Arc Length**

If a curve C is described by the parametric equations
$x = f(t)$,$y = g(t)$,$\alpha \leqslant t \leqslant \beta$, where $f'$ and $g'$ are continuous on $[\alpha, \beta]$ and C is traversed exactly once as t increases from$\alpha$ to $\beta$, then the length of C is

$$L = \int_{\alpha}^{\beta} \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2} dt \qquad ds$$

**Surface Area**

In the same way as for arc length, we can adapt to obtain a formula for surface area. If the curve given by the parametric equations
$x = f(t)$,$y = g(t)$,$\alpha \leqslant t \leqslant \beta$ is rotated about the x-axis, where $f'$ and $g'$ are continuous and $g(t) \geqslant 0$, then the area of the resulting surface is given by

$$S = \int_{\alpha}^{\beta} 2\pi y \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2} dt \qquad ds$$

# 7 Polar Coordinates

## 7.1 Slope

$$x = r\cos\theta, \quad y = r\sin\theta$$

where r can be regarded as a function of $\theta$. Hence we have

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$$

## 7.2 Arclength and Surface Area

Suppose we have the function in polar coordinates:

$$r = f(\theta), \quad a \leq \theta \leq b$$

For the area enclosed by this function, we have:

$$A = \int_a^b \frac{1}{2} f^2(\theta) d\theta$$

For the arclength enclosed by this function, we have:

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$